



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

Bachelor of Science in Computer Science and Engineering Program

Final Examination, Fall 2024 Semester

Course: CSE489 Mobile Programming Section-01
Instructor: Dr. Md Mostofa Kamal Rasel, Associate Professor.
Full Marks: 25 (25 will be counted)
Time: 1 Hour 5 Minutes
Notes: Answer all questions. The marks of each question are listed on the right.

1. Use Android Asynchronous Task instead of ExecutorService in the following program.

[CO2,C2
Marks: 6]

```
public void fileDownloadTask(Context c, String src, String dest,
                             TextView ptv, ProgressBar pb){
    Handler handler = new Handler(Looper.getMainLooper());
    ExecutorService executor = Executors.newSingleThreadExecutor();
    executor.execute() -> {
        URL url = new URL(src);
        HttpURLConnection huc = (HttpURLConnection) url.openConnection();
        huc.connect();
        long fileSize = huc.getContentLength();
        int bufCapacity = 1024 * 256;
        byte[] buffer = new byte[bufCapacity];
        int chunkSize;
        long downloaded = 0;
        InputStream is = huc.getInputStream();
        FileOutputStream fos = c.openFileOutput(dest, Context.MODE_PUBLIC);
        while((chunkSize = is.read(buffer)) > 0){
            downloaded += chunkSize;
            double progress = Math.round(downloaded * 100 / fileSize, 2);
            handler.post() -> {
                ptv.setText(progress+"%");
                pb.setProgress(progress);
            };
            fos.write(buffer, 0, chunkSize);
        }
        fos.close();
        huc.disconnect();
        handler.post() -> {
            Toast.makeText(c, "Completed", Toast.LENGTH_SHORT).show();
        };
    };
}
```

2. Suppose that you have used a browser app to read an online newspaper at 07:00 AM, opened the Facebook app at 08:00 AM, checked the upcoming events in the event management app at 07:30 AM, and briefly read another online newspaper on the browser app at 10:05 AM and 03:05 PM. Meanwhile, you have used the attendance app for the courses CSE489 and CSE406, which ended at 11:40 AM and 04:40 PM, respectively. Now, you are searching for a video on the YouTube app by pausing a mobile game that you were playing and downloading a movie in the background using a background service at the same time. Given that your device does not have enough free memory, **prepare** the list of victim applications in a sequence of priority that is followed by the Android system to keep the gaming application running if the YouTube app needs additional memory.
- [CO2,C2
Marks: 2]

3. *QuizBank* is an MCQ question bank app that offers a vast library of multiple-choice questions to practice, learn, and prepare for exams. Perfect for students and professionals, QuizBank provides customizable quizzes to address topic-based questions, real-time performance tracking, a leaderboard to check ranks and offline access. With a user-friendly interface and daily challenges, QuizBank helps build confidence, expand knowledge, and boost test scores. [CO3,C2,C3 Marks: 3+5]

Given that *QuizBank* stores data either locally or remotely or in both places based on their kind, **suggest** your preferred technique (i.e. Own Processing Technique, API, or Webview) *for each case* of implementation, authentication & verification, optimization, and data security.

Write down *XML* and *Java* code to load and show the website <http://www.quizbank.info> using a Webview.

4. SholoGutee is a turn-based board game played by two persons. This game is widely popular in sub-continental countries such as Bangladesh, India, and Nepal. Suppose you are going to develop an Android application to play SholoGutee. Your developed gaming application must allow the users to play SholoGutee offline or online. [CO3,C2 Marks: 3]
Now, **discuss** which business model would be most beneficial for you as a developer.
5. In a turn-based mobile game application, such as SholoGutee, the opponent player is informed about the game board status either after a countdown is finished or after every move that a player takes. Suppose that informing the opponent player is done by triggering an event called *turnCompleted()*. Given that the countdown cannot be done in the UI thread, either *Asynchronouns Task* or *Service* must be implemented to handle the countdown. [CO3,C2,C3 Marks: 2+4]
Now, **discuss** the type of service you prefer to use for the countdown and **write** down the necessary declarations in *XML* and *Java* code for your service, which are required to notify the Activity that the countdown is completed.